A271 – Landauer and the Phase-Space Accounting

Johann Pascher

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.1 Use of AI Tools

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.2 Question and Answer

The widespread misreading. Landauer's principle is commonly read as follows: erasing one bit costs at least $k_B T \ln 2$ — because information has energy, because a bit carries a physical minimum price, because entropy and information are the same thing. From this it follows: the more information is processed, the more energy it costs; whoever computes frugally pays less; whoever erases a lot pays more. This reading is standard in large parts of computer science, cognitive science, and popular physics.

What Landauer actually wrote [1]. The decisive sentence appears in § 1, before any entropy calculation begins:

> *"In most instances a computer pushes information around in a manner that is* independent of the exact data *which are being handled, and is only a function of the* physical circuit connections.*"*

[1], § 1

And the naive energy argument — "a degree of freedom carries kT of thermal energy, therefore a bit costs kT" — he dismisses immediately afterwards himself:

> *"This argument does* not make it clear *that the signal energy must actually be dissipated."*

[1], § 1

The thermodynamic accounting depends on the *circuit connections* — the wiring, the construction of the computer — not on the data, not on the program, not on the content of the bits. No energy can be assigned to an information bit; energy dissipates in the hardware element that is clocked, because the wiring switches it. That is Landauer's actual statement.

Why it is so often misread. Landauer writes for engineers who know computer architecture. "Degree of freedom associated with the information" means in technical English: the component that serves to store information. Anyone reading without that background hears: information has a degree of freedom, therefore an energy. That is precisely the misunderstanding Landauer wanted to refute.

[B] The central conclusion. The Landauer formula yields no energy quantity for information — not even when applied to the physical bit. It gives the price of a region reduction of a *concrete component* under *concrete conditions* (quasi-static, uniform distribution, two-state system with specific switching kinetics). Which component, which kinetics, which conditions — that is determined by the *physical circuit connections*, not by the bit. The same logical operation on different hardware costs different amounts. The qubit shows: even "one bit" is not an unambiguous physical unit. The formula describes physics — but not the physics of information; the physics of hardware elements that are clocked.

What this document does. It exposes the phase-space accounting behind Landauer's argument — region reduction, Liouville, bath as volume receiver — and shows at every step: the accounting counts regions of physical elements; it reads no content.

[STIPULATION] A physical system occupies a region in phase space. This region has a size — measurable in units of hf (Section .7). It exists independently of how one names, subdivides, or describes it. That is the physical starting point. Everything else comes afterwards.

[B] Descriptions come from outside. If one decides to divide the region into two subregions and call them "A" and "B", with target region "Z", one has introduced a description of the region reduction. Calling them instead "0", "1" and "0" (traditional notation) creates the confusion with logical bits — this convention obscures that "0" designates both the source region and the target region, and suggests an information content where there is none. If one divides more finely — into 1000 zones — one has introduced approximately 10 logical bits. If one assigns only a number between 0 and 1, one has introduced an analogue value. Physics forces none of these decisions. The region is the same physical region in all cases.

The terms "bit", "byte", "qubit", "analogue value" are labels for description decisions — not for physical objects. They subdivide the region; they do not create it.

[B] The Landauer bit is hardware — not information. This is regularly confused. Clarification in one paragraph:

A Landauer bit is a physical region with two stable macro-states — a transistor, a capacitor, a wire that sits high or low. A bus bit is the same: a physical region with two stable macro-states, embedded in a parallel hardware architecture. A memory cell is the same. The three are structurally identical. Thermodynamics treats all three equally: $Q \geq k_B \cdot T \cdot \ln(W_{A+B}/W_Z)$ — regardless of whether the region is called "Landauer bit", "bus bit", or "memory cell".

The hardware is fixed before any computation. It does not change through what one computes with it. A 64-bit bus has 64 physical regions — before the computation, during the computation, after the computation. What the computation does with these regions — which mappings it executes, which regions are reduced to which target regions — is a logical description of these physical processes. The description does not exist in the hardware; it is an external interpretation.

The information content of a bit — whether it means "true" or "false", whether it belongs to a correct computation step, whether it is read at all — is a decision at the description level. Thermodynamics does not know this level. It only knows: region A and region B exist; after the region reduction, target region Z exists; $W_Z \leq W_{A+B}$; the bath absorbs the difference.

[B] Three levels — strictly separated:

Level 1 — Construction. The engineer decides during chip design which region operations exist — bijective or non-bijective. This is cast in silicon, invariant during operation, before any computation. This decision fixes the thermodynamic cost structure — not the computation, not the program.

Level 2 — State transitions during operation. Once the computer runs, state transitions occur clock-cycle by clock-cycle — uniformly, at the clock frequency, regardless of what is being computed. Landauer costs arise only for *non-bijective* region mappings — when a component reduces multiple input regions to one target region Z through its physical construction. This is fixed by the hardware architecture, not by any computation operation. A transistor flipping from region A to region B executes a bijective mapping — no volume loss, no Landauer costs. What the description level does afterwards with the component's state is thermodynamically irrelevant.

The Landauer costs run at clock frequency, practically constant, as long as the computer runs — because the non-bijective region mappings are fixed by the designer and occur with every clock cycle. CPU utilisation says little about this: it measures how many clock cycles are used for computations at the description level — but at idle, clock transitions continue to run through. The thermodynamic difference between high and low utilisation is small, as long as no physical parts are actually switched off. Clock gating, power gating — those are hardware decisions at Level 1, not program decisions. The hardware determines — not the program. Landauer states this explicitly in § 1 [1]:

> "*In most instances a computer pushes information around in a manner that is* independent of the exact data *which are being handled, and is only a function of the* physical circuit connections."

[1], § 1

The clocking couples the elements to the bath and sets the switching kinetics in motion — but it does not change which elements are bijectively or non-bijectively connected. That is fixed by the wiring, before every clock cycle, independent of what is being computed.

[B] From this it follows directly: no energy can be assigned to an information bit — only to the hardware element that is clocked. The energy dissipates because the element switches, not because the bit carries a particular content, has a truth value, or is read at all. The element switches because the wiring prescribes it — independent of the data. The thermodynamic accounting is blind to content; it knows only the element and its switching state.

[K] Why the paper is so often misunderstood. Landauer writes from the perspective of an engineer who understands computer architecture. He immediately dismisses the naive energy argument — "a binary device must have at least one degree of freedom associated with the information; classically a degree of freedom is associated with kT of thermal energy" — himself:

> "*This argument does* not make it clear *that the signal energy must actually be dissipated.*"

[1], § 1

And he establishes that his Class-1 devices can hold information *without* dissipation — dissipation arises only *when switching* [1], § 2. Anyone with a background in computer engineering reads this immediately correctly: the degree of freedom belongs to the physical component, not to the bit as an abstract object. Anyone reading without that background confuses "degree of freedom associated with the information" with an energy assignment to the information itself — and arrives at precisely the misunderstanding that Landauer was trying to refute.

Level 3 — Description. The program, the computation, an AI controller can decide which hardware is used — switch parts off, choose routes, determine that the full bit-width is not needed for a given result. This has absolutely nothing to do with Landauer. It is resource management at the description level. The thermodynamic costs of the hardware used still arise — fixed by construction and switching moment, regardless of what the description level does with the results.

[K] How much is inside a physical region? A colloidal particle in a 1-µm trap contains physically around $7.9 \cdot 10^9$ orthogonal quantum states (`landauer_pruefskript.py`, Check 7). Calling two zones roughly "left" and "right" describes this region with 1 logical bit. Dividing it into 1000 zones describes the same region with ≈ 10 logical bits. The physical region is identical in both cases — the description depth is an observer's choice.

[B] The cost of erasure depends on the region, not on the description. The formula is:

$$Q \geq k_B \cdot T \cdot \ln\left(\frac{W_{\text{before}}}{W_{\text{after}}}\right) \tag{1}$$

W_{before} and W_{after} are physical region sizes in units of hf. How one names this ratio — whether as "erasing 1 bit" or "factor-2 reduction" — changes nothing in the physics.

[K] The analogue case makes this particularly clear — and at the same time shows its complexity. How many distinguishable states an analogue signal contains depends on several factors, all physical: the bandwidth B, the temperature T and the resistance R (Johnson–Nyquist noise: $U_{\text{rms}} = \sqrt{4k_BTRB}$ [18]), the sensitivity of the measurement technique, the signal-to-noise ratio of the concrete system, and further external sources. There is no universal resolution limit for analogue signals — it is a system property, not a natural constant.

What can be said in general: when the resolution limit is set by the thermal noise of the bath itself (the physically deepest case), the principle already formulated in Section .7 applies: the same bath that collects the erasure costs limits the grading of the analogue signal. The region count $W = L/\delta x$ and the erasure costs $k_BT \ln W$ scale with the same temperature — the accounting is self-consistent. How many logical bits one assigns to the analogue region is independent of this and remains a description decision (Check 8: illustration with declared δx).

[B] Shannon and Boltzmann are different bookkeeping systems [16, 17]. Shannon describes how much uncertainty a description carries: $H = -\sum_i p_i \log_2 p_i$ — a quantity about descriptions. Boltzmann describes how large a physical region is: $S = k_B \ln W$ — a quantity about regions. The two are identical only when the description maps exactly onto the physical micro-states and these are uniformly distributed. In any other case — coarse division, unequal occupation, continuous systems — they are different things with different units.

[K] [K] What Landauer actually said in 1961 — in the original:

> *"It is argued that computing machines inevitably involve devices which perform logical functions that do not have a single-valued inverse. This logical irreversibility is associated with physical irreversibility and requires a minimal heat generation, per machine cycle, typically of the order of kT for each irreversible function."*

[1], Abstract

And in § 3 he specifies:

> *"We shall call a device* logically irreversible *if the output of a device does not uniquely define the inputs."*

[1], § 3

Landauer speaks throughout of *devices* — physical hardware components — and of *logical functions* as their operations. He means a hardware property: a mapping in which different physical input states lead to the same physical output state — a many-to-one mapping of the states of the information-bearing system. This is a statement about the *physical structure of the operation itself*, not about an external description decision. Landauer does not draw this separation: he understands logical irreversibility as a physical property of the hardware — bijective or non-bijective — that exists regardless of how one names or interprets the states. Logical irreversibility is a *sufficient* condition for heat production in a certain class of operations — not the only one. Every physically irreversible process produces heat, regardless of whether it is described as "logically irreversible"; that is exactly the point of Section .11.**[B]** The error of the standard presentation is thus: it treats the description unit — the bit — as if it were a physical minimum size. It is not. A physical minimum size does exist: hf. But it has nothing to do with bits. The region comes first; how many bits one ascribes to it is a subsequent decision.

.3 The One Idea from Which Everything Follows

A bit is physical: two *distinguishable* states of a system. Two wells for a particle, two magnetisation directions, two voltage levels. What matters is only: the system can be in state "0" or "1", and one can tell them apart.

Region reduction: Regardless of which region — A or B — the system was in before, afterwards it is in target region Z. The target region Z is freely choosable; what counts is exclusively the ratio W_{A+B}/W_Z.

$$\begin{array}{ccc} \text{A} & \longrightarrow & \\ & \searrow & \text{Z} \\ \text{B} & \longrightarrow & \end{array}$$

Two regions are reduced to one — the target region Z. This mapping is *not reversible*: knowing only the target region, one cannot reconstruct which source region the system was in. That — and nothing else — generates the cost (Section .5).

.4 What "Phase Space" Means Here

Phase space is the space of all possible micro-states (all positions and momenta of all particles). A macro-state such as "region A" is not a point in this space but a *region*: many micro-states (thermal jitter in the left well — each jitter position counts as "0").

[STIPULATION] The Boltzmann entropy measures this region:

$$S = k_B \cdot \ln W \tag{2}$$

W = phase-space volume of the region. Entropy here is a **measure of the size of the occupied region** — nothing mysterious.

Phase space before:		Phase space after:
A | B	$\Longrightarrow$	Z | empty
Volume: $2W_0$ $(A+B)$		Volume: W_0 (only Z)

[B] Erasure halves the accessible volume:

$$\Delta S_{\text{region}} = -k_B \cdot \ln 2 \tag{3}$$

Pure volume accounting: ΔS follows solely from the ratio of the physical region sizes W_0 — the same result for every transistor, every capacitor, every other hardware that executes a region reduction. (Generalisation to N levels and non-uniform distributions: Section .7.)

.5 Why It Is Not Free: Liouville

The microscopic dynamics (classically Hamiltonian, quantum-mechanically unitary) has a hard property, **Liouville's theorem** [2]:

[B] The time evolution of a closed system *preserves the phase-space volume*. Deform, shift, stir — yes. Shrink — never.

Clarification: Liouville holds exactly only for the closed system *without* a bath. For the sub-system "bit" alone — after coupling to the bath — it no longer holds. What is closed is only the total system bit + bath; for this, the total volume is preserved, while the bit sub-region may shrink at the expense of the bath region.

Intuitively: the phase-space flow is an *incompressible fluid*. An ink drop in water can be deformed arbitrarily; its volume remains.

The trap closes:

- Erasure *requires*: $2W_0 \to W_0$ (halving).
- Liouville *forbids*: volume shrinkage in the closed system.

[B] Consequence: a closed system cannot erase. Not an engineering problem — a theorem. The only way out: the system must not be closed. The volume must go somewhere — into the bath (Section .6).

.6 The Bath as Volume Receiver

When the bit is coupled to a heat bath, Liouville holds for the *total system* bit + bath. If the bit region is to shrink by half (uniform case: both regions equally large), the bath region must at least *double*. In the general case $\Delta S_{\text{bath}} \geq -\Delta S_{\text{bit}}$ (second law); the generalisation to $\ln(W_{\text{before}}/W_{\text{after}})$ follows in Section .7:

$$\Delta S_{\text{bath}} \geq +k_B \cdot \ln 2 \tag{4}$$

A bath at temperature T enlarges its region by absorbing energy; the fundamental relation $\mathrm{d}S = \delta Q/T$ translates this into heat:

[B]

$$Q \geq T \cdot \Delta S_{\text{bath}} = k_B \cdot T \cdot \ln 2 \tag{5}$$

This is the **Landauer bound** [1, 3]:

Liouville (volume preserved) + accounting (bit region halved) ⇒ bath region doubled ⇒ heat $\geq k_B T \ln 2$ into the bath.

At 300 K: $k_B T \ln 2 \approx 2.87 \cdot 10^{-21}$ J per bit (Check 1).

Two conditions directly readable:

- **No erasure without a bath** — there is no other volume receiver.
- **Equality only quasi-statically** — any finite speed generates additional dissipation $W \approx k_B T \ln 2 + B/\tau$ [4, 5].

.7 Why 2? — The Counting Basis

Binariness is *convention, not physics*. In general:

$$Q \geq k_B \cdot T \cdot \ln\left(\frac{W_{\text{before}}}{W_{\text{after}}}\right) \tag{6}$$

— a ratio of region sizes. N levels reduced to one: $\ln N$ (Check 4). The $\ln 2$ is the computer-science special case.

[B] What counts as a level? Not energy. Energy levels are not a valid counting basis: they can be degenerate or have unequal spacings without changing the count. The criterion is *distinguishability*: two states count separately when they can in principle be told apart without error. For *quantum systems* this means orthogonality in Hilbert space; for *classical systems* it means distinguishability within the thermal resolution limit set by the bath. A bit can lie in two degenerate states of equal energy — ideally even: holding costs nothing then, only forgetting. Level spacings act only indirectly, via thermal occupation probabilities. The proper currency is therefore the Gibbs form [6]:

$$S = -k_B \cdot \sum_i p_i \ln p_i \tag{7}$$

with $\ln N$ as the special case of uniform distribution. Unequally spaced, unequally occupied systems automatically carry less entropy (Checks 5–6).

[B] The fundamental discretisation exists — and it is hf. Classically the phase-space volume has units; W is defined only up to an arbitrary cell size. The accounting is undisturbed by this, since only differences enter — the cell cancels. Quantum mechanics sets the cell absolutely: the number of orthogonal states in a region Γ of a system with f degrees of freedom is

$$N_{\text{max}} = \Gamma/h^f \quad [7] \tag{8}$$

(Check 7: colloid in 1-μm trap $\approx 7.7 \cdot 10^9$ orthogonal positional states in the 1D position space — the "2" of the experiment is a coarse choice, not fundamental.)

The quasi-analogue case. A value in range L with resolution δx carries $\ln(L/\delta x)$ levels; region reduction costs $k_B T \ln(L/\delta x)$ (Check 8). δx is not free: the bath noise sets it — the same bath that collects the erasure costs limits the grading. No analogue trick bypasses the bound.

Table 1: Fundamental vs. emergent two-valuedness

	Spin-$\frac{1}{2}$	Colloid in double well
Origin of the 2	fundamental: Hilbert space $\mathbb{C}^2$	emergent: 2 metastable macro-regions
Validity	always	only while barrier $\gg k_B T$
below that	nothing	quasi-continuum of micro-states

.8 Why the Content Does Not Matter

The entire calculation has at *no point* asked what meaning region A or B carries, whether the bit belongs to a correct computation or not, whether it is interpreted as "true" or "false". Only this entered: *how many* distinguishable regions before, *how many* after.

Thermodynamics counts regions; it does not read them. Region A and region B are physically equivalent — physics knows no preferred direction, and the target region Z is freely choosable.

[B] Region reduction costs identically for every assignment — regardless of how one interprets the regions. There is no thermodynamic content or truth detector.

In the same language: a non-terminating computation dissipates energy without bound and has no truth value during this process. Energy turnover and semantic validity lie on different levels — phase-space accounting here, representation there. Category separation: physical validity (hardware level) vs. representational validity (content level).

.9 The Same Statement from the Hardware Side

The statement from Section .8 — thermodynamics reads no content — can be derived independently of the accounting directly from the physical structure of the computer.

[B] The computer does not shrink. A 64-bit processor has the same number of transistors, the same number of wires, the same number of degrees of freedom before and after every operation. The phase space of the hardware remains constant — 2^{64} possible states before, 2^{64} after. The result may need only 1 bit; the apparatus that produced it is nevertheless just as large.

[B] What actually goes into the bath is the correlation. The logical mapping is many-to-one: many different input states land on the same output. Landauer calls this in § 4 [1] the physical *many-into-one* mapping and states:

> *"Hence the physical 'many into one' mapping, which is the source of the entropy change, need not happen in full detail during the machine cycle which performed the logical function. But it must eventually take place, and this is all that is relevant for the heat generation argument."*

[1], § 4

The language of correlations comes from Bennett [3]: the 63 discarded bits are logically gone, but physically the transistors — the elements in Landauer's sense — still carry the trace of which input states set them; only this trace is no longer reconstructible from the result. Landauer calls this [1], Summary, the *"unnecessary details of the computer's past history"* — meaning the states of the elements that led to the result, not a temporal sequence. Precisely what is non-reconstructible goes as heat to the lattice. Not the silicon, but the elements that have forgotten which inputs they had.

[K] The entropy accounting alone is not sufficient — Landauer needs the dynamics as the vehicle. In the Summary [1] he states this explicitly:

> *"The information-bearing degrees of freedom of a computer* interact *with the thermal reservoir represented by the remaining degrees of freedom. This interaction plays two roles. First of all, it acts as a sink for the energy dissipation involved in the computation. [...] Secondly, the interaction acts as a source of noise causing errors."*

[1], Summary

The coupling of the information-bearing degrees of freedom to the heat bath is the physical mechanism — without switching kinetics there is no heat flow, regardless of what the region accounting says. That is why Landauer devotes § 5 to a detailed analysis of the switching kinetics: relaxation time τ_0, switching time τ_s, Boltzmann error. Thermodynamics gives the lower bound; dynamics is the vehicle that enforces it.

[B] The forgetting process is content-neutral. A processor computing $2+2=4$ and one computing $2+2=5$ switch the same number of transistors, discard the same bit-width, and pay the same thermodynamic bill to the bath. Physics does not distinguish the two. A result in region A costs the same as one in region B — physics does not distinguish the regions.

.10 Hardware and Computation Are Different Things

A fundamental error in the standard Landauer presentation: the hardware is treated as an image of the computation — as if the physical apparatus only exists and is only thermodynamically active when a logical operation is being executed. This is wrong.

[B] The hardware is not the computation. A 64-bit processor is a physical system with a 64-bit state space. This state space exists regardless of whether a meaningful operation is running, whether the result needs 1 bit or 64 bits, what meaning the result carries, whether anyone reads the result or not. The hardware is not an abstraction — it is silicon, wires, fields, capacitors, continuously interacting with their environment.

[B] The computation is an abstraction over the hardware. It describes what the programmer or mathematician sees — input, output, function. The hardware does not know this abstraction. It knows only state transitions, charge displacements, heat dissipation.

The consequence for Landauer. When one asks "what does erasing a bit cost", one takes the abstraction as the starting point and looks for the thermodynamic costs of the logical operation. That is a legitimate question — but it misses the physics of the real system. The real computer produces entropy continuously, independent of the logical abstraction currently running on it.

.11 The Computer Always Computes

A physical computer does not stop computing when it waits for a result. The clock frequency runs, the transistors switch, the power supply delivers current — continuously, regardless of whether "useful" work is happening. An idle processor runs empty loops. A waiting server sends keepalive packets. An operating system polls interrupts. There is no state "computer is not computing" while it is switched on.

This means: the forgetting process from Section .10 runs *always*. The thermodynamic bill — correlation loss, heat to the lattice — does not stop, regardless of whether the result is needed or not.

This sharpens Landauer's formulation. He isolates a single discrete event — "region reduction costs $k_B T \ln 2$". The reality is a *continuous stream* of state transitions, correlation losses, and heat dissipations, 3 billion times per second in a modern processor, without pause.

Concrete example: Dynamic RAM. A DRAM bit is a capacitor holding charge. But capacitors lose charge through leakage currents — continuously, unstoppably. Therefore every DRAM cell must be *refreshed* every 32–64 milliseconds: the refresh operation reads the state and writes it back. This happens clock-cycle by clock-cycle, automatically, for each of the billions of cells — regardless of whether the data are currently needed, what meaning they carry, whether the program is sleeping.

This is not a side effect — it is the physical nature of the system. The DRAM refresh is *not* a region reduction in the Landauer sense; it is a maintenance operation, and its direct costs are primarily engineering costs (leakage currents, recharging). But the reason it must happen at all is thermodynamic: the capacitor couples permanently to the heat bath.

But that is not the whole point. The refresh *itself* is not a passive reaction — it requires active computation: a counter must run, the address of the next row to refresh must be computed, a timing signal must be generated, the memory bus must be locked, the sense amplifier must be activated. That is real computation — with real state transitions, real correlation losses, real entropy flow into the bath. And this happens 15,000 times per second, for each memory bank, without interruption.

The computer pays entropy to exist — not only to compute. A DRAM that "does nothing" for one hour has nonetheless undergone $\approx 112{,}000$ refresh cycles per cell during that hour and produced entropy continuously. The stored information was the same throughout — but the thermodynamic bill ran through.

.12 Quantum Computers: Decoherence as Continuous Entropy Flow

For the quantum computer the same principle applies as for DRAM — the physical apparatus continuously fights thermodynamic decay — but the physics is more relentless.

What decoherence is. A qubit in superposition has a phase relationship to its environment. Every smallest interaction — a phonon, a photon, a magnetic field fluctuation — changes this phase. The qubit does not lose its quantum nature because information is destroyed, but because the correlation between qubit and environment is distributed into the bath. Again Liouville: the volume is preserved, but it distributes over qubit-plus-environment correlations that no one controls any more.

The aggravation compared to DRAM. For the classical DRAM one can refresh — read state, write back, costs thermodynamically, works, because 0 and 1 are robust. For the qubit this is in principle impossible: a measurement collapses the superposition. An unknown quantum state can neither be copied (no-cloning theorem [12]) nor measured without changing it. Refresh in the classical sense does not exist.

What does exist: *quantum error correction* [13]. But this is thermodynamically even more expensive than DRAM refresh — it needs 100–1000 physical qubits per logical qubit, continuous syndrome measurements and classical correction operations. All of this runs through, clock-cycle by clock-cycle, for each logical qubit, even when no gate operation is being executed.

Decoherence time as DRAM analogue. A superconducting qubit typically has $T_2 \approx 100\ \mu$s [14] — after that, the phase information is distributed in the bath. That is seven times shorter than a DRAM refresh interval (64 ms), but qualitatively different: DRAM loses information slowly and steadily, the qubit loses it exponentially fast.

Content-neutral as always. Decoherence does not distinguish between a qubit carrying a correct quantum state and one carrying a different region. It destroys both equally fast. The thermodynamic price of quantum computing does not depend on the content — it depends on the physics of the system.

[H] Open question. Whether there is a minimal price in principle for maintaining quantum coherence — independent of the quality of error correction — is an open question of quantum thermodynamics [15].

.13 Region Operations — Volume-Preserving or Volume-Reducing

Every physical state change is a **mapping between phase-space regions**. The thermodynamically decisive question is exclusively: does the mapping preserve the region volume, or does it reduce it? Whether this mapping belongs to a "computation", whether it is described as a logical operation, whether anyone considers it meaningful — that is completely irrelevant for thermodynamics.

[B] Volume-preserving mappings — the region is reshaped, not reduced. Liouville holds. No thermodynamic lower bound. Traditionally such mappings are called "reversible gates" (NOT, CNOT, Toffoli) — physically they are *bijective region mappings*. Whether they are part of a computation or not is irrelevant.

[B] Volume-reducing mappings — multiple input regions land on the same target region Z. Volume shrinks. The bath must absorb the difference. Traditionally called "irreversible gates" (AND, OR, ERASE) — physically *non-bijective region mappings*. Again: whether part of a computation or not is irrelevant.

[B] Bennett [3] showed that any sequence of region mappings can in principle be executed as a pure sequence of volume-preserving mappings — intermediate states are carried along and reversibly released at the end. The thermodynamic costs arise not at the intermediate steps, but exclusively at the final relinquishment of regions:

[B] The lower bound attaches not to the mapping, but to the relinquishment of regions — to the forgetting.

.14 The Accounting in a Table

Table 2: Phase-space accounting for region reduction

Step	Bit region	Bath region	Total (Liouville)
before	$2W_0$	W_{bath}	$2W_0 \cdot W_{\text{bath}}$
reduction to Z	$W_Z \leq W_0$	?	must $\geq 2W_0 W_{\text{bath}}$ remain
therefore	W_Z	$\geq 2W_{\text{bath}}$	preserved ✓
price	$\Delta S = -k_B \ln(W_v/W_Z)$	$\Delta S \geq +k_B \ln(W_v/W_Z)$	$Q \geq k_B T \ln(W_v/W_Z)$

.15 The Experiments and Their Limits

The image. Two glasses stand on a table — region A and region B. Next to them, the ocean. Region reduction means: one glass is poured into the ocean. The glass volume does not disappear — it disperses across 1.4 billion cubic kilometres. Liouville is satisfied: not a drop is lost.

But: which glass it was — A or B — is distributed so finely in the ocean's noise that no real instrument can ever separate it out again. Not because the information is destroyed (it is encoded in every current, every temperature gradient). But because the resolution limit of every real observer lies ten orders of magnitude below the signal.

And: the ocean cannot afterwards under the same external conditions provide the same representation capacity for the two glasses. It can no longer show them separately — the target region Z is smaller than $A + B$. That is not ignorance; it is a physical fact about region sizes.

The price of pouring: $Q \geq k_B T \ln 2$. Not due to ignorance — because the ocean had to absorb the volume and heat flowed for this.

(Check 2–3): Real CMOS switching event $\approx 10^{-17}$–10^{-16} J — four to five orders of magnitude above Landauer. All bit erasures in the world together would have Landauer costs in the watt-to-kilowatt range; data centres actually consume tens of gigawatts [8] — factor 10^7–10^{10}. In every real machine the bound is invisible; it is the natural-law lower bound, the rest is engineering loss (CV^2-recharging, leakage currents, data transport).

[K] The lower bound is universal — but how far the real dissipation lies above it depends fundamentally on the physical realisation of the bit. This is not a weakness of the theory but Landauer's own statement [1], Summary:

> *"Actual devices which are far from minimal in size and operate at high speeds will be likely to require a much larger energy dissipation to serve the purpose of erasing the unnecessary details of the computer's past history."*

[1], Summary

Three examples show the range:

- **Single spin** — one single degree of freedom, coupling to the phonon bath via spin-lattice relaxation, barrier set by Zeeman splitting. The realisation is closest to the bound.
- **DRAM capacitor** — $\sim 10^{10}$ electrons jointly realise one bit. Leakage current is continuous; refresh costs dominate the Landauer bound by many orders of magnitude. The realisation determines the distance to the bound.
- **Magnetic domain** — collective state of $\sim 10^6$ spins. Domain wall motion as switching mechanism. The bound $k_B T \ln 2$ holds as a lower limit; actual costs depend on domain geometry and wall mobility.

The claim "$k_B T \ln 2$ per bit" describes the *lower bound* of a single region reduction in the quasi-static limit — not the real dissipation of any concrete system. How much a real bit costs depends on the wiring, geometry, and switching kinetics of the component — exactly what Landauer in § 1 calls the *physical circuit connections*.

[H] The actual lower bound is hard to pin down. The $k_B T \ln 2$ holds for the classical two-state region reduction — but whether that is the smallest possible physical realisation of a bit remains open. The qubit demonstrates this: it uses the same physical degree of freedom as a classical spin ($\mathscr{H} = \mathbb{C}^2$), but carries a continuous complex amplitude — more than 1 classical bit in the same physical structure. What counts as the "minimal realisation of a bit" depends on how one counts bits — and that is a description decision, not a physical boundary. Whether there is an absolute thermodynamic lower bound *below* $k_B T \ln 2$ is an open question.

[K] The ocean image. The bit volume that tips into the bath during region reduction stands in such an extreme disproportion to the bath (1 degree of freedom against 10^{23}) that it lies *below the resolution limit* of every real instrument. The bath absorbs the volume — Liouville is satisfied — but no measuring instrument can separate the signal from the bath's noise any more.

[B] Epistemic vs. ontological — a necessary caveat. Not measurable does not mean non-existent. The micro-states of the bath after the region reduction all still exist; every particle has a precise position and momentum; the information about which source region the system was in still resides in the correlations — Liouville guarantees this explicitly. The bath has absorbed the volume, not destroyed it.

Landauer holds *not because* information is ontologically destroyed. It holds *because* the bath must absorb the volume (Liouville) — that is a statement about the mechanics, not about knowledge or perception.

[K] Maxwell's demon [11] and Bennett's resolution [3]. A being knowing all 10^{23} micro-states of the bath — a Laplace demon — could trace the bit back in thought. Bennett's resolution 1982: the demon must *erase its measurement notes* to begin a new cycle — and that costs exactly $k_B T \ln 2$. The actual site of entropy production is the erasure of the notes, not the measurement itself.

The single-particle experiments: Bérut et al. 2012 [4] (colloid in optical double well, cycle times 10–40 s), Jun/Gavrilov/Bechhoefer 2014 [5] (feedback trap, higher precision). Both approach the bound and confirm the $1/\tau$ form of the additional dissipation (Check 9).

Three declared scepticism points:

1. **Extrapolation**: what is confirmed is a limit ($\tau \to \infty$) that no experiment enters; the bound is fitted as an asymptote.
2. **Signal $\approx$ noise**: $k_B T \ln 2$ lies in the order of magnitude of the thermal fluctuation of a single measurement; the result exists only as an ensemble average.
3. **Success rate**: the protocols erase with 90–95 % reliability; the accounting of failed erasures enters the balance.

Feedback extension: with measurement + feedback the generalised second law $W_{\text{ext}} \leq -\Delta F + k_B T I$ (Sagawa–Ueda [9]), experimentally confirmed [10, **?**]. Information is a resource in the accounting with energy exchange rate $k_B T$ per nat (Check 10).

.16 Connection to Open Questions (Internal)

- **Feedback accounting in fast systems**: computation operations at ns time scales lie ten orders of magnitude from the tested quasi-static regime; there the B/τ term dominates everything. Anyone wanting to measure a Landauer-like residual in fast recurrent systems must include the Sagawa–Ueda term $k_B T I$ [9] explicitly in the conventional budget before a residual can count as new — otherwise one measures feedback thermodynamics, not new physics.
- **FFGFT contact**: none. The Landauer accounting lives at the level of statistical mechanics; the ξ-scheme is not touched (its own level, its own accounting — P20/P39-analogue).

.17 Summary in Five Sentences

Region reduction shrinks the phase-space region of the system; Liouville forbids volume shrinkage in closed systems; therefore a bath must absorb the volume, and the price is heat $\geq k_B T \ln(W_v/W_n)$ — the $\ln 2$ only as the binary special case. Counting is by distinguishability (orthogonality), not by energy; the absolute fundamental discretisation yields hf, everything coarser is emergent counting at a chosen level. Not measurable does not mean non-existent: the micro-states of the bath all still exist, the information about the erased bit resides in correlations — Landauer holds because of the mechanics (Liouville), not because of ignorance; the irreversibility is statistical, not ontologically absolute. The accounting counts regions and reads no contents — every assignment costs identically. In practice the bound is never reached, because any finite speed generates additional dissipation that dominates by orders of magnitude.

.18 Layer Status Summary

Table 3: Layer status of statements

Statement	Status
Bit ≠ physical region (Shannon vs. Boltzmann)	[B]
$S = k_B \ln W$ as starting point	[STIPULATION]
Region reduction $\{A, B\} \to Z$	[STIPULATION]
Region reduction $\Rightarrow \Delta S = -k_B \ln(W_{A+B}/W_Z)$	[B]
Liouville theorem	[B]
Closed system cannot erase	[B]
Landauer bound $Q \geq k_B T \ln 2$	[B]
Numerical value $2.87 \cdot 10^{-21}$ J at 300 K	[K]
Counting basis = orthogonality, not energy	[B]
hf as fundamental discretisation	[B]
Gibbs form as general entropy	[STIPULATION]
Content does not enter the accounting	[B]
Hardware ≠ computation	[B]
Computer does not shrink	[B]
Correlation into bath (not silicon)	[B]
Forgetting process content-neutral	[B]
DRAM 112,000 refreshes/h	[K]
No-cloning theorem	[B]
$T_2 \approx 100\ \mu$s superconducting qubits	[K]
Minimal price for coherence maintenance	[H]
Volume-preserving mappings: no lower bound	[B]
Volume-reducing mappings: bath pays	[B]
Epistemic vs. ontological: correlations remain	[B]
Sagawa–Ueda term $k_B T I$	[K]

15× [B] · 8× [K] · 3× [STIPULATION] · 1× [H]

.19 Verification Scripts

- `python/A_Serie_Skripte/a270_landauer.py` (Checks 1–10)

.20 Open Questions

First. Whether there is a minimal price in principle for maintaining quantum coherence — independent of the quality of error correction — is an open question of quantum thermodynamics [H].

Second. The question of the continuous entropy flow of a physical computer as a function of its clock frequency, its bit-width, and its logical irreversibility is not conclusively treated in the literature.

.21 Supersedes

This document has no direct legacy counterpart; it is a standalone working document on Landauer accounting and phase-space analysis.

.22 References

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[7] Semi-classical state counting Γ/h^f: standard; e.g. Landau/Lifshitz, *Statistical Physics*, § 7.

[8] Data centre energy consumption: IEA reports 2024/25, order of magnitude several hundred TWh/year worldwide ($\approx$ tens of GW continuous power). Figure serves only for order-of-magnitude comparison.

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[14] F. Arute et al. (Google AI Quantum), *Quantum supremacy using a programmable superconducting processor*, Nature **574**, 505 (2019) — typical T_2 times of superconducting qubits $\approx 100\ \mu$s.

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